Discrete Mathematics – Final Exam

2024 – Term B

Instructions for students:

1. In the exam there are **8** short questions and **3** open questions.
2. You must answer **all** questions (No choice).
3. The questions are not ordered according to difficulty!
4. Answers should be written on the exam form.
5. The notebooks can be used as drafts but will not be graded.
6. When asked to write explicitly a "set/function/relation," you must specify all the elements of the set/function/relation.
7. Use of any external aiding material is **forbidden**. For your convenience, a formula sheet is attached at the end of the form.
8. The duration of the exam is **three hours.**

Good luck!

Part A – Short Questions

- Write your answers clearly in the marked boxes.
- No need to explain how you got to the final answer, **unless it is explicitly asked**.
- In counting questions, you can use any of the convenient notations which we used in class, **except** for the symbol $\sum$ and three dots, '…'.

Question 1 (6 points)

Given the following propositions, write the numbers of the propositions that are **not** equivalent to: $\neg q \to (p \vee \neg s)$.

1. $\neg s \to (p \vee q)$
2. $\neg p \to \neg(s \to q)$
3. $(\neg p \wedge s) \to q$
4. $q \wedge p \wedge \neg s$

1,2,4

Question 2 *(6 Points)*

Disprove: *If* $(A \backslash B) \cap (A \backslash C) = \emptyset$ *then* $A \subseteq B \cap C$.

$A = \{1\}, B = \{1,2\}, C = \{3\}$

Question 3 *(6 points)*

Write an explicite function $f: P(\mathbb{Z} \backslash \{5\}) \to P(\mathbb{Z}) \backslash P(\{5\})$ that is not 1-1 (injective).

$\forall A \in P(\mathbb{Z} \backslash \{5\}): \quad f(A) = \{1\}$

Question 4 *(6 points)*

A race involves 8 runners wearing shirts of different colors: red, pink, orange, yellow, green, blue, purple and black. The following facts are known:

- The runner in the red shirt finished the race before the runner in the pink shirt.
- The runner in the orange shirt finished before the runner in the yellow shirt.
- The runner in the green shirt finished before the runner in the blue shirt.
- The runner in the purple shirt finished before the runner with the black shirt.

What is the number of possible outcomes of the race?

Assume that no two runners arrive at the same time.

$$\binom{8}{2}\cdot\binom{6}{2}\cdot\binom{4}{2}\cdot\binom{2}{2}=\binom{8}{2,2,2,2}$$

Question 5 *(6 points)*

What is the number of different sequences of natural numbers of length 20, that satisfy the following conditions:

1. The first number in the sequence is 0 and the last number is m where *m<20*.
2. Each number in the sequence is either equal to the previous number or greater than it by *exactly* one.

$$\binom{19}{m}$$

We need to choose m out of the 19 "jumps" between adjacent slots in which the next element is increased by 1, while the other 19-m jumps will be zero (the next element is identical).

Question 6 *(6 points)*

How many integer solutions to, $x_1+x_2+x_3+x_4+x_5+x_6=15$, *that satisfy:*

$\forall i:\ 0\le x_i\le 5$ *?*

$$\binom{15+6-1}{5} - \binom{6}{1}\binom{9+6-1}{5} + \binom{6}{2}\binom{3+6-1}{5}$$

Question 7 *(6 points)*

Let $\mathrm{A} = \{0,1,2,\dots,999\}$. Define an <u>equivalence relation</u> R on A as follows.

xRy if and only if the set of the digits composing the number x equals the set of digits composing y.

For example, $23R332$ since the set of digits composing both numbers is $\{2,3\}$.

Additionally, $5R555$ since the set of digits composing both numbers is $\{5\}$.

However, $\neg(450R454)$ since the sets $\{4,5,0\},\{4,5\}$ are different.

What is the number of equivalence classes of *R*?

$$\binom{10}{1} + \binom{10}{2} + \binom{10}{3}$$

All subsets of {1,2,…, 9} of sizes one two and three (for every such subset, there is an equivalent class containing all numbers composed of these digits).

Question 8 *(6 points)*

Give an example of a relation R on a non-empty set A such that R is not symmetric, not transitive, but satisfies the following property*:*

$$\forall x,y,z \in A\ \big((xRy \wedge yRz) \rightarrow (x=z)\big)$$

$$A = \{1,2,3\}$$

$$R = \{(1,2),(2,1),(3,4)\}$$

Part B – Open Questions

- *Write your answers clearly in the marked boxes. You must prove your answers.*
- ***Write clear and readable answers! Your grade will also depend on the clarity of your proof and not only the correctness of your answer!***

Question 9 *(16 points)*

Prove by induction that for every $n \geq 12$, there exist two natural numbers $k, l \in \mathbb{N}$ such that $n = 4k + 5l$.

By induction on n.

Base case: for n=12, it holds that $n = 12 = 3 \cdot 4 + 0 \cdot 5$.

Step: Assume the claim holds for $12 \leq n$, that is, $\exists k', l' \; s.t \quad n = 4k' + 5l'$

and we next show correctness for $n + 1$.

By the induction hypothesis, $\exists k', l' \; s.t \quad n = 4k' + 5l'$. We consider the following cases:

Case 1: $k' \neq 0$: In this case we can write n+1 as:

$n + 1 = 4k' + 5l' + 1 = 4(k' - 1) + 5(l' + 1)$. Since $k' \neq 0$, $(k' - 1) \in \mathbb{N}$, and the claim follows for $k = k' = 1, l = l' + 1$.

Case 2: $k' = 0$: In this case $n = 5l'$. Since $n \geq 12$, we can conclude that $l' \geq 3$.

We can then write n+1 as: $n + 1 = 5l' + 1 = 5(l' - 3) + 15 + 1 = 5(l' - 3) + 4 \cdot 4$.

Since $l' \geq 3$, $it\ holds\ that\ (l' - 3) \in \mathbb{N}$ and the claim follows for $l = (l' - 3) and\ k = 4$.

Question 10 (20 points)

Given two functions $f, g: \mathbb{N} \to \mathbb{N}$. It is known that for every $n \in \mathbb{N}$ it holds that $f(n) = 2g(n) - 1$.

A. Prove that f is not onto

B. Prove that if f is 1-1 (injective), then g is also 1-1.

A. Consider the number $0 \in \mathbb{N}$. We will show that 0 has no pre-image in f.

Assume towards contradiction there exists $n \in \mathbb{N}$ s.t $f(n) = 0$. By the definition of f it holds that $f(n) = 2g(n) - 1 = 0$, and therefore $g(n) = \frac{1}{2} \notin \mathbb{N}$, in contradiction to the fact that the co-domain of g is $\mathbb{N}$.

B. Assume that f is 1-1 and we will next show g is 1-1. Let $n_1, n_2 \in \mathbb{N}$ such that $g(n_1) = g(n_2)$. It then follows that $2g(n_1) = 2g(n_2)$ and also $2g(n_1) - 1 = 2g(n_2) - 1$.

Thus, by the definition of f it holds that $f(n_1) = f(n_2)$.

Since f is 1-1 we can then conclude that $n_1 = n_2$.

Question 11 (16 points)

For each of the following relations, determine whether it is an equivalence relation, weak partial order relation, strong partial order relation or none of them. Prove your answer.

A. R is a relation on $\mathbb{N} \times (\mathbb{N} \setminus \{0\})$ where $(a,b)R(c,d)$ iff $a \cdot d = b \cdot c$

B. S is a relation on $\mathbb{N} \times (\mathbb{N} \setminus \{0\})$ where $(a,b)S(c,d)$ iff $a \cdot c = b \cdot d$

A. R is an equivalence relation.

Reflexive: Let $(\mathrm{a},\mathrm{b}) \in \mathbb{N} \times \mathbb{N} \setminus \{0\}$. Since $\mathrm{a} \cdot \mathrm{b} = \mathrm{b} \cdot \mathrm{a}$, by the definition of R $(\mathrm{a},\mathrm{b})\mathrm{R}(\mathrm{a},\mathrm{b})$.

Symmetric: For every two pairs (a,b) , $(\mathrm{c},\mathrm{d}) \in \mathbb{N} \times \mathbb{N} \setminus \{0\}$, if $(\mathrm{a},\mathrm{b})\mathrm{R}(\mathrm{c},\mathrm{d})$, $\mathrm{a} \cdot \mathrm{d} = \mathrm{b} \cdot \mathrm{c}$, since multiplicity is commutative it also holds that $\mathrm{c} \cdot \mathrm{b} = \mathrm{d} \cdot \mathrm{a}$ and therefore $(\mathrm{c},\mathrm{d})\mathrm{R}(\mathrm{a},\mathrm{b})$.

Transitive: Let $(\mathrm{a},\mathrm{b}), (\mathrm{c},\mathrm{d}), (\mathrm{e},\mathrm{f}) \in \mathbb{N} \times \mathbb{N} \setminus \{0\}$, such that $(\mathrm{a},\mathrm{b})\mathrm{R}(\mathrm{c},\mathrm{d})$ and $(\mathrm{c},\mathrm{d})\mathrm{R}(\mathrm{e},\mathrm{f})$.

By the definition of R, $\mathrm{a} \cdot \mathrm{d} = \mathrm{b} \cdot \mathrm{c}$ and $\mathrm{c} \cdot \mathrm{f} = \mathrm{d} \cdot \mathrm{e}$. Because $\mathrm{b}, \mathrm{d}, \mathrm{f} \neq 0$, we can conclude that $\frac{\mathrm{a}}{\mathrm{b}} = \frac{\mathrm{c}}{\mathrm{d}}$, and $\frac{\mathrm{c}}{\mathrm{d}} = \frac{\mathrm{e}}{\mathrm{f}}$. Therefore $\frac{\mathrm{a}}{\mathrm{b}} = \frac{\mathrm{e}}{\mathrm{f}} \rightarrow \mathrm{a} \cdot \mathrm{f} = \mathrm{b} \cdot \mathrm{e}$, and by the definition of R, $(\mathrm{a},\mathrm{b})\mathrm{R}(\mathrm{e},\mathrm{f})$.

B. R is neither of them.

The relation R is not an equivalence relation, and also not a weak partial order relation because it is not reflexive: for example consider the ordered pair (1,2). Since $1 \cdot 1 \neq 2 \cdot 2$ it holds that $\big((1,2),(1,2)\big) \notin \mathrm{R}$.

It is also not a strong partial order relation because it's not asymmetric. For example, $(1,2)\mathrm{R}(2,1)$ and $(2,1)\mathrm{R}(1,2)$.

(it is also not anti-reflexive as (1,1)R(1,1)).

Extra page

Discrete Mathematics – Formula sheet for the exam

Set Theory:

$$A\Delta B = (A\backslash B) \cup (B\backslash A)$$
$$A \times B = \{(a,b)|\ a \in A\ and\ b \in B\}$$
$$P(A) = \{S|S \subseteq A\}$$

$$|A \times B| = |A| \cdot |B|$$
$$|P(A)| = 2^{|A|}$$

De-Morgan:

$$\overline{B \cup C} = \bar{B} \cap \bar{C} \qquad \mathrm{A}\backslash(B \cup C) = (A\backslash B) \cap (A\backslash C)$$
$$\overline{B \cap C} = \bar{B} \cup \bar{C} \qquad \mathrm{A}\backslash(B \cap C) = (A\backslash B) \cup (A\backslash C)$$

Distributive:

$$\mathrm{A} \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$\mathrm{A} \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Functions/Relations:

$$i_{\mathrm{A}} = \{(a,a)|a \in A\}$$

Logic:

For all formulas α, β, γ :

De-Morgan:

$$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$$
$$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$$

Distributive:

$$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$$
$$\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$$

Negation rules:

$$\neg\forall \mathrm{x}(\alpha) \equiv \exists x(\neg\alpha)$$
$$\neg\exists \mathrm{x}(\alpha) \equiv \forall x(\neg\alpha)$$

Combinatorics:

Choosing k items out of n different items:

	With repetition	Without repetition
With order	n^k	$\frac{n!}{(n-k)!}$
Without order	$\binom{k+n-1}{n-1}$	$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!}$

Binomial Formula

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot x^k y^{n-k}$$

Multinomial Coefficient:

$$\binom{n}{i_1, i_2, i_3, \dots, i_k} = \frac{n!}{i_1! \cdot i_2! \cdot i_3! \cdot \dots \cdot i_k!} \quad if \;\; i_1 + i_2 + i_3 + \cdots + i_k = n$$

Inclusion-exclusion principle:

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq \{1,2,\dots,n\},\;\; I \neq \emptyset} (-1)^{|I|-1} \cdot \left| \bigcap_{\mathrm{i} \in \mathrm{I}} \mathrm{A_i} \right|$$